Money in Flexible Price Environments: MIU

Extra Notes for Dynare Code

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Readings

- Walsh, C. E., 2010. Monetary Theory and Policy, 3rd Ed. The MIT Press. (CH 2)
- Sidrauski, M., 1967. Rational choice and patterns of growth in a monetary economy. *American Economic Review*, 57(2): 534-544.

The Model

Firm

Aggregate output is produced according to the following constant returns to scale technology:

$$y_t = f(k_{t-1}, n_t, z_t) = z_t k_{t-1}^{\alpha} n_t^{1-\alpha} \tag{1}$$

where the technology shock z_t follows:

$$lnz_t = \rho_z lnz_{t-1} + \epsilon_{z,t}, \qquad 0 < \rho_z < 1, \qquad \epsilon_{z,t} \sim i.i.d.(0, \sigma_z^2) \tag{2}$$

and the capital accumulation equation:

$$k_t = (1 - \delta)k_{t-1} + x_t \tag{3}$$

The Model

Household

Household maximizes the discounted expected utility function, where utility is nonseparable in consumption and real money balances:

$$E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, m_t, 1-n_t) = E_0 \sum_{t=0}^{\infty} \beta^t \{ \frac{[ac_t^{1-\eta_m} + (1-a)m_t^{1-\eta_m}]^{\frac{1-\eta_c}{1-\eta_m}}}{1-\eta_c} + \frac{(1-n_t)^{1-\eta_n}}{1-\eta_n} \} \quad (4)$$

s.t. the budget constraint:

$$Y_t + T_t + (1-\delta)K_{t-1} + \frac{(1+i_{t-1})B_{t-1}}{P_t} + \frac{M_{t-1}}{P_t} = C_t + K_t + \frac{B_t}{P_t} + \frac{M_t}{P_t} \quad (5)$$

and in real per capita term:

$$y_t + \tau_t + (1-\delta)k_{t-1} + \frac{(1+i_{t-1})b_{t-1}}{1+\pi_t} + \frac{m_{t-1}}{1+\pi_t} = c_t + k_t + m_t + b_t \quad (6)$$

The Model

Government budget constraint

$$T_t = M_t - M_{t-1} \tag{7}$$

Money creation process

- Money growth

$$M_t = (1 + \theta_t) M_{t-1} \tag{8}$$

- Money supply shock

$$\mu_t = \rho_m \mu_{t-1} + \phi z_{t-1} + \epsilon_{m,t} \quad \epsilon_{m,t} \sim i.i.d.(0, \sigma_m^2) \tag{9}$$

where $\rho_m > 0$ and

$$\mu_t = \theta_t - \bar{\theta} \tag{10}$$

Solving the Model

Household's decision is characterized by the following value function (control variables $\{c_t, n_t, k_t, m_t, b_t\}$):

$$V(k_{t-1}, m_{t-1}, b_{t-1}) = \max[U(c_t, m_t, 1 - n_t) + \beta E_t V(k_t, m_t, b_t)] \tag{11}$$

s.t.

$$\begin{aligned} c_t &= y_t + \tau_t + (1-\delta)k_{t-1} + \frac{(1+i_{t-1})b_{t-1}}{1+\pi_t} + \frac{m_{t-1}}{1+\pi_t} \\ &\quad -k_t - m_t - b_t \\ y_t &= f(k_{t-1}, n_t, z_t) = z_t k_{t-1}^{\alpha} n_t^{1-\alpha} \end{aligned}$$

Solving the Model

FOCs w.r.t. $\{k_t, m_t, b_t, n_t\}$:

$$1 = \beta E_t\Big\{ \frac{U_c(c_{t+1}, m_{t+1}, 1-n_{t+1})}{U_c(c_t, m_t, 1-n_t)} [f_k(k_t, n_{t+1}, z_{t+1}) + (1-\delta)]\Big\} \quad (12)$$

$$U_c(c_t, m_t, 1-n_t) = U_m(c_t, m_t, 1-n_t) + \beta E_t\Big(\frac{U_c(c_{t+1}, m_{t+1}, 1-n_{t+1})}{1+\pi_{t+1}}\Big) \quad (13)$$

$$1 = \beta E_t\Big[\frac{U_c(c_{t+1}, m_{t+1}, 1-n_{t+1})}{U_c(c_t, m_t, 1-n_t)}\Big(\frac{(1+i_t)}{1+\pi_{t+1}}\Big)\Big] \quad (14)$$

$$f_n(k_{t-1}, n_t, z_t) = \frac{U_n(c_t, m_t, 1-n_t)}{U_c(c_t, m_t, 1-n_t)} \quad (15)$$

Solving the Model

Remarks:

□ (12) is the Lucas asset pricing equation for physical assets (capital).

□ (13) states that the MU_c at t must equal to the marginal benefit of adding to money holdings at t, which consists of two parts: 1. money directly yields utility (U_m) in t; 2. indirectly through the discounted higher future consumptions in real term (Walsh 2003: 49).

□ The FOC for n_t, (15) states that the MP_n, $f_n(k_{t-1}, n_t, z_t)$ is equal to the MRS between leisure $(1 - n_t)$ and consumption.

Solving the Model

Remarks:

□ Using Fisher equation to rewrite the FOC for b_t, (14)

$$1 = \beta E_t\Big[\frac{U_c(c_{t+1}, m_{t+1}, 1-n_{t+1})}{U_c(c_t, m_t, 1-n_t)} (\frac{1+i_t}{1+\pi_{t+1}})\Big] \tag{16}$$

$$\frac{1}{1+r_t} = \beta E_t\Big[\frac{U_c(c_{t+1}, m_{t+1}, 1-n_{t+1})}{U_c(c_t, m_t, 1-n_t)}\Big]. \tag{17}$$

Substitute (17) into (13):

$$\frac{U_m(c_t, m_t, 1-n_t)}{U_c(c_t, m_t, 1-n_t)} = 1 - (\frac{1}{1+r_t}) E_t(\frac{1}{1+\pi_{t+1}}) \tag{18}$$

$$\frac{U_m(c_t, m_t, 1-n_t)}{U_c(c_t, m_t, 1-n_t)} = \frac{i_t}{1+i_t} \tag{19}$$

(18) states that the MRS between money and consumption is equal to the price, or the opportunity cost, of holding money, which depends on both r_t and $E_t\pi_{t+1}$.

Solving the Model

Remarks:

□ The welfare costs of inflation

In the MIU model, money holdings yield direct utility and higher inflation reduces real money balances (from previous slide), $\Rightarrow$ inflation generates a welfare loss as long as $i_t > 0$. So, what is the optimal rate of inflation?

Since,

- private opportunity cost of holding money is i_t;
- social marginal cost of producing money is 0;

Therefore, if $i_t = 0$, this inefficiency would be eliminated and the optimal rate of inflation is reached.

So, what is the optimal rate of inflation?

Function Forms

Separable vs. nonseparable utility function:

Recall the utility function (4):

$$E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, m_t, 1-n_t) = E_0 \sum_{t=0}^{\infty} \beta^t \Big\{ \frac{[a c_t^{1-\eta_m} + (1-a) m_t^{1-\eta_m}]^{\frac{1-\eta_c}{1-\eta_m}}}{1-\eta_c} + \frac{(1-n_t)^{1-\eta_n}}{1-\eta_n} \Big\}$$

- Separable utility ($\eta_c = \eta_m$):

In the special case that $\eta_c = \eta_m = 1$, preferences over consumption and real balances are log linear and the utility function (4) becomes:

$$E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, m_t, 1-n_t) = E_0 \sum_{t=0}^{\infty} \beta^t [a \ln(c_t) + (1-a) \ln(m_t) + \frac{(1-n_t)^{1-\eta_n}}{1-\eta_n}] \tag{20}$$

If the utility is separable in consumption and real balances, the MIU model exhibits 'superneutrality' property, which we will discuss in details later.

Function Forms

- Separable vs. nonseparable utility function:
  - Nonseparable utility ($\eta_c \neq \eta_m$):

Define $\tilde{C}_t = a c_t^{1-\eta_m} + (1-a) m_t^{1-\eta_m}$

$$U_c(c_t, m_t, 1-n_t) = a\tilde{C}_t^{\frac{\eta_m-\eta_c}{1-\eta_m}} c_t^{-\eta_m} \quad (21)$$

$$U_m(c_t, m_t, 1-n_t) = (1-a)\tilde{C}_t^{\frac{\eta_m-\eta_c}{1-\eta_m}} m_t^{-\eta_m} \quad (22)$$

and the $MRS_{m,c}$:

$$\frac{U_m(c_t, m_t, 1-n_t)}{U_c(c_t, m_t, 1-n_t)} = \left(\frac{1-a}{a}\right)\left(\frac{m_t}{c_t}\right)^{-\eta_m} \quad (23)$$

Function Forms

□ Separable vs. nonseparable utility function:

- Nonseparable utility ($\eta_c \neq \eta_m$):

Taking the derivative of U_c with respect to m_t, we have:

$$U_{cm} = a\tilde{C}_t^{\frac{\eta_m - \eta_c}{1-\eta_m} - 1} c_t^{-\eta_m}(\eta_m - \eta_c)(1-a)m_t^{-\eta_m} \ . \qquad (24)$$

Therefore, if $\eta_m > \eta_c$, $\Rightarrow U_{cm} > 0$. In other words, consumption and real balances are complements.

And the greater η_m, the greater the complementarity between consumption and real balances.[1]

[1] $d(ln(m_t/c_t))/d(lnMRS_{m,c}) = -1/\eta_m$

Function Forms

☐ Money demand function:

Combine (23) and (19), we have:

$$\left(\frac{1-a}{a}\right)\left(\frac{m_t}{c_t}\right)^{-\eta_m} = \frac{i_t}{1+i_t} \tag{25}$$

$$m_t = c_t\left(\frac{a}{1-a}\right)^{-\frac{1}{\eta_m}}\left(\frac{i_t}{1+i_t}\right)^{-\frac{1}{\eta_m}} \tag{26}$$

The linearized (26) gives us the standard money demand function:

$$\hat{m}_t = \hat{c}_t - \left(\frac{1}{\eta_m}\right)\left(\frac{1}{i}\right)\hat{i}_t \tag{27}$$

The real demand for money is a negative function of the nominal interest rate and a positive function of consumption. The coefficient $1/\eta_m$ is the magnitude of the elasticity with respect to the opportunity cost $i/(1+i)$. The elasticity with respect to i is $-1/[\eta_m(1+i)]$; with rate deviations, the slides and code use the small-rate approximation $-1/(\eta_m i^{ss})$.

Preferences, growth and the code

Let $Q = [ac^{1-\eta_m} + (1-a)m^{1-\eta_m}]^{1/(1-\eta_m)}$. Walsh (2010), Eq. 2.64, uses

$$U = \frac{Q^{1-\eta_c}}{1-\eta_c} + \Psi \frac{(1-n)^{1-\eta_n}}{1-\eta_n}.$$

- Code mapping: $\Phi = \eta_c$, $b = \eta_m$, $\eta = \eta_n$; a weights consumption. At $\eta_n = 1$, use $\Psi \log(1-n)$.
- Along balanced growth with c, m, w proportional to a trend A, wU_c scales as $A^{1-\eta_c}$, while leisure utility is unchanged at fixed hours. Constant hours therefore require $\eta_c = 1$ under these additive preferences.
- This restriction concerns trend growth, not a stationary business-cycle model. Walsh abstracts from growth; so do these code files.

The simpler utility and Appendix 2.7

- Additive CRRA consumption and money utility with hours disutility has the same growth restriction: $wc^{-\eta_c}$ scales as $A^{1-\eta_c}$. It also removes consumption–money complementarity.
- A growth-compatible alternative is $U = [Qv(1-n)]^{1-\eta_c}/(1-\eta_c)$. The wage-weighted consumption marginal utility and the marginal cost of labour then share the trend factor; at $\eta_c = 1$ its limit is $\log Q + \log v(1-n)$.
- Appendix 2.7, p. 85: Walsh's Ω_1 equals $-\texttt{omega_one}$; his $\Omega_2 = \texttt{omega_two}$. Our Euler equation substitutes out marginal utility.
- Hatted quantities are log deviations; r, i and π are level deviations of net rates. The code follows Walsh's small-rate approximations, rather than exact gross-rate scaling.
- The code fixes `nss`; its implicit Ψ supports that steady state. Changing trend inflation at fixed `nss` does not implement the slide's fixed-preference steady-state comparison.

Steady States

Four steady state values from the Cooley-Hansen CIA model are applicable to this MIU model:

$$R^{ss} = \frac{1}{\beta} \tag{28}$$

$$\frac{y^{ss}}{k^{ss}} = \frac{1}{\alpha}(R^{ss} - 1 + \delta) \tag{29}$$

$$\frac{c^{ss}}{k^{ss}} = \frac{y^{ss}}{k^{ss}} - \delta \tag{30}$$

$$\frac{n^{ss}}{k^{ss}} = (\frac{y^{ss}}{k^{ss}})^{\frac{1}{1-\alpha}} \tag{31}$$

In addition to these, in steady state we have:

$$\frac{m^{ss}}{c^{ss}} = (\frac{a}{1-a})^{-\frac{1}{\eta m}}(\frac{1+\pi^{ss}-\beta}{1+\pi^{ss}})^{-\frac{1}{\eta m}} \tag{32}$$

$$\frac{(n^{ss})^{\eta c}}{(1-n^{ss})^{\eta n}} = a^{\frac{1-\eta c}{1-\eta m}}(1-\alpha)(\frac{y^{ss}}{k^{ss}})^{\frac{-\alpha}{1-\alpha}}(\frac{c^{ss}}{k^{ss}})^{-\eta c}(\frac{k^{ss}}{n^{ss}})^{-\eta c}$$

$$[1+(\frac{a}{1-a})^{\frac{-1}{\eta m}}(\frac{1+\pi^{ss}-\beta}{1+\pi^{ss}})^{\frac{\eta m-1}{\eta m}}]^{\frac{\eta m-\eta c}{1-\eta m}} \tag{33}$$

Steady States

Remarks:

- The MIU model ((28) -(31)) exhibits neutrality property: proportional changes in the level of nominal money balances do not affect real variables;
- The MIU model ((28) -(31)) exhibits superneutrality property: the ratios in these four equations are invariant to money growth; full superneutrality additionally requires invariant labour, as when $\eta_c = \eta_m$;
- (33) states that, if $\eta_m - \eta_c > 0$ (money and consumption are complements), $\Rightarrow \pi^{ss}(\theta^{ss}) \uparrow$, $i^{ss} \uparrow$, which will result the RHS of (33) to fall. Therefore $n^{ss} \downarrow$, so are the y^{ss}, c^{ss}, and k^{ss}. Intuitively, Higher inflation reduces real money balances and decreases marginal utility of consumption. Households will substitute toward leisure and to decrease labor supply [2].
  If $\eta_m = \eta_c$, n^{ss} is independent of money growth. However, m^{ss}/c^{ss} still depends on the growth rate of nominal money.

[2] If c and m are substitutes, the effects will be opposite.

Dynamics

The linearized model:

$$\hat{y}_t = \alpha \hat{k}_{t-1} + (1-\alpha)\hat{n}_t + \hat{z}_t \tag{34}$$

$$(\frac{y^{ss}}{k^{ss}})\hat{y}_t = (\frac{c^{ss}}{k^{ss}})\hat{c}_t + \hat{k}_t - (1-\delta)\hat{k}_{t-1} \tag{35}$$

$$\hat{r}_t = \alpha(\frac{y^{ss}}{k^{ss}})(E_t\hat{y}_{t+1} - \hat{k}_t) \tag{36}$$

$$\Omega_1 \hat{c}_t = \Omega_1 E_t \hat{c}_{t+1} - [\hat{r}_t + \Omega_2(E_t\hat{m}_{t+1} - \hat{m}_t)] \tag{37}$$

$$\hat{y}_t - (\Omega_1\hat{c}_t - \Omega_2\hat{m}_t) = (1+\eta_n\frac{n^{ss}}{1-n^{ss}})\hat{n}_t{}^3 \tag{38}$$

$$\hat{i}_t = \hat{r}_t + E_t\hat{\pi}_{t+1} \tag{39}$$

$$\hat{m}_t = \hat{c}_t - (\frac{1}{\eta_m})(\frac{1}{i})\hat{i}_t \tag{40}$$

$$\hat{m}_t = \hat{m}_{t-1} - \hat{\pi}_t + \mu_t \tag{41}$$

$$\hat{z}_t = \rho_z\hat{z}_{t-1} + \epsilon_{z,t} \tag{42}$$

$$\hat{\mu}_t = \rho_\mu\hat{\mu}_{t-1} + \phi_z\hat{z}_{t-1} + \epsilon_{\mu,t} \tag{43}$$

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[3] $\Omega_1 = \gamma\eta_c + (1-\gamma)\eta_m$, $\Omega_2 = (\eta_m - \eta_c)(1-\gamma)$, and $\gamma = [1 + a^{-1}(1-a)(m^{ss}/c^{ss})^{1-\eta_m}]^{-1}$.

Dynamics

Remarks

- (37), (38), and (40) are different from those in the CIA model. In the case in which the utility is separable in consumption and real balances ($\eta_c = \eta_m$), (37) & (38) become:

$$\eta_c \hat{c}_t = \eta_c E_t \hat{c}_{t+1} - \hat{r}_t \quad (44)$$

$$\hat{y}_t - \eta_c \hat{c}_t = (1 + \eta_n \frac{n^{ss}}{1 - n^{ss}}) \hat{n}_t \quad (45)$$

  which are the standard linearized FOCs for consumption and labour.

- The linearized system now (in the case of $\eta_m = \eta_c$) can be separated into three independent blocks . . .

Dynamics

Remarks

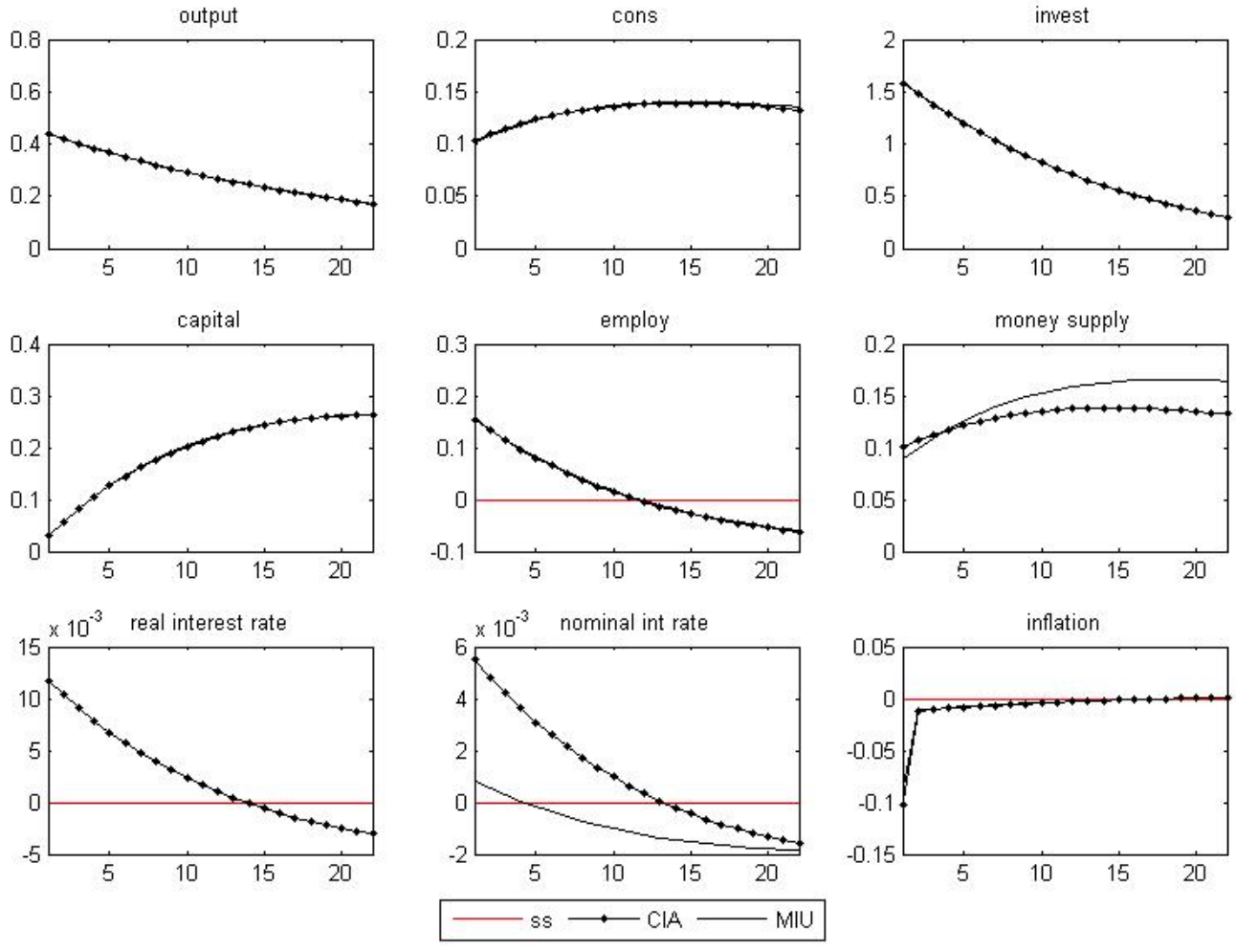
1. The first block includes the first five equations, which shows that the dynamics of $\hat{y}, \hat{c}, \hat{r}, \hat{k}, \hat{n}$ can be solved independently of the money growth (inflation).
2. The second block consists of equation (39)$-$(41), which shows that inflation and real money balances can be solved independently of the real side of the model, the first block.

   $\Leftarrow$ **Once again, the MIU model exhibits neutrality and superneutrality!**

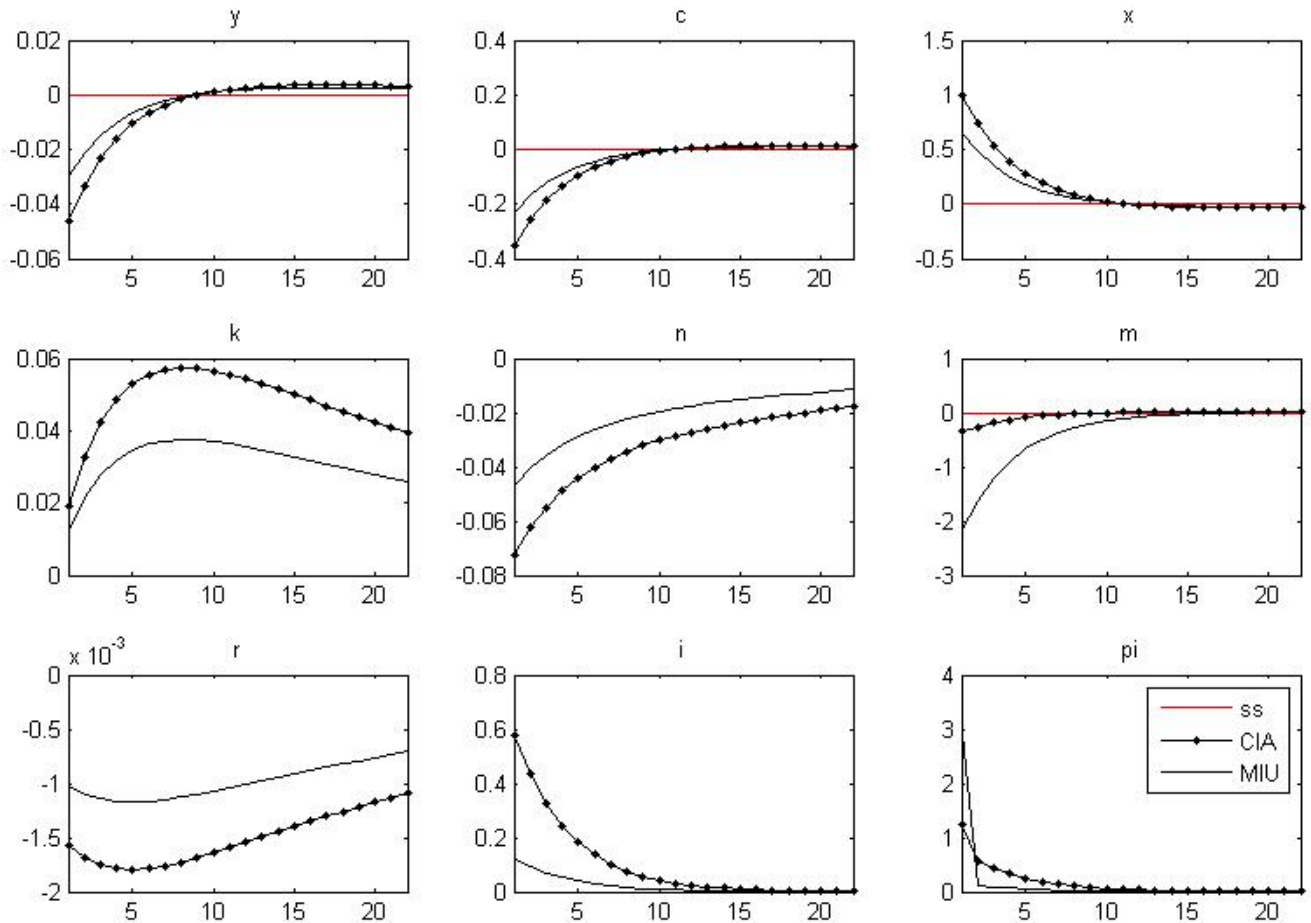
3. The last block consists of the disturbances of the model.

- Finally, since real balances is one argument of households' utility, the money demand (40) can be derived explicitly from households' optimization problem, not in an ad hoc manner.

Dynamics (Impulse response to a technology shock)



Dynamics (Impulse response to a money supply shock)



4

[4] To match CIA, share of money in utility 9 times larger. Contrary to calibration of 9 times smaller. This only has a visible impact on real variables; m , r , i, pi unaffected. Do you know why?

Appendix

Appendix A: Steady States

- $\frac{m^{ss}}{c^{ss}}$

From (25) we have:

$$\frac{m_t}{c_t} = (\frac{a}{1-a})^{-\frac{1}{\eta m}} (\frac{i_t}{1+i_t})^{-\frac{1}{\eta m}} \tag{A.1}$$

Since $1 + i^{ss} = R^{ss}(1 + \pi^{ss})$ and $R^{ss} = \frac{1}{\beta}$, in steady state:

$$(\frac{i^{ss}}{1+i^{ss}}) = (\frac{1+\pi^{ss}-\beta}{1+\pi^{ss}}) \tag{A.2}$$

Substitute (A.2) into (A.1),

$$\frac{m^{ss}}{c^{ss}} = (\frac{a}{1-a})^{-\frac{1}{\eta m}} (\frac{1+\pi^{ss}-\beta}{1+\pi^{ss}})^{-\frac{1}{\eta m}} \tag{A.3}$$

Appendix A: Steady States

- $\tilde{C}^{ss}$

In steady state,

$$\tilde{C}^{ss} = a(c^{ss})^{1-\eta_m} + (1-a)(m^{ss})^{1-\eta_m} \quad \text{(A.4)}$$

$$\tilde{C}^{ss} = a(c^{ss})^{1-\eta_m} + (c^{ss})^{1-\eta_m}(1-a)(\frac{m^{ss}}{c^{ss}})^{1-\eta_m} \quad \text{(A.5)}$$

Substitute (A.3) into (A.5),

$$\tilde{C}^{ss} = a(c^{ss})^{1-\eta_m} + (c^{ss})^{1-\eta_m}(1-a)[(\frac{a}{1-a})^{-\frac{1-\eta_m}{\eta_m}}(\frac{1+\pi^{ss}-\beta}{1+\pi^{ss}})^{-\frac{1-\eta_m}{\eta_m}}] \quad \text{(A.6)}$$

$$\tilde{C}^{ss} = a(c^{ss})^{1-\eta_m}[1 + (\frac{1-a}{a})(\frac{a}{1-a})^{\frac{\eta_m-1}{\eta_m}}(\frac{1+\pi^{ss}-\beta}{1+\pi^{ss}})^{\frac{\eta_m-1}{\eta_m}}] \quad \text{(A.7)}$$

$$\tilde{C}^{ss} = a(c^{ss})^{1-\eta_m}[1 + (\frac{a}{1-a})^{\frac{-1}{\eta_m}}(\frac{1+\pi^{ss}-\beta}{1+\pi^{ss}})^{\frac{\eta_m-1}{\eta_m}}] \quad \text{(A.8)}$$

Appendix A: Steady States

- n^{ss}

From the FOC with respect to n_t (15), we know:

$$U_\ell(c_t, m_t, 1-n_t) = U_c(c_t, m_t, 1-n_t) f_n(k_{t-1}, n_t, z_t) \quad \text{(A.9)}$$

$$\frac{(1-n_t)^{-\eta_n}}{a c_t^{-\eta_m} \tilde{C}_t^{\frac{\eta_m-\eta_c}{1-\eta_m}}} = (1-\alpha)\frac{y_t}{n_t} \quad \text{(A.10)}$$

In steady state, we have:

$$(1-n^{ss})^{-\eta_n} = a(c^{ss})^{-\eta_m}(1-\alpha)\left(\frac{y^{ss}}{n^{ss}}\right)(\tilde{C}^{ss})^{\frac{\eta_m-\eta_c}{1-\eta_m}} \quad \text{(A.11)}$$

Appendix A: Steady States

- n^{ss}

Making use of the steady state values of $\tilde{C}^{ss}$, $\frac{y^{ss}}{k^{ss}}$, and $\frac{n^{ss}}{k^{ss}}$,

$$
\begin{aligned}
(1-n^{ss})^{-\eta_n} &= a(c^{ss})^{-\eta_m}(1-\alpha)\left(\frac{y^{ss}}{k^{ss}}\right)^{\frac{-\alpha}{1-\alpha}} \\
&\quad \{a(c^{ss})^{1-\eta_m}\left[1+\left(\frac{a}{1-a}\right)^{\frac{-1}{\eta_m}}\left(\frac{1+\pi^{ss}-\beta}{1+\pi^{ss}}\right)^{\frac{\eta_m-1}{\eta_m}}\right]\}^{\frac{\eta_m-\eta_c}{1-\eta_m}} \qquad \text{(A.12)} \\
(1-n^{ss})^{-\eta_n} &= a^{\frac{1-\eta_c}{1-\eta_m}}(c^{ss})^{-\eta_c}(1-\alpha)\left(\frac{y^{ss}}{k^{ss}}\right)^{\frac{-\alpha}{1-\alpha}} \\
&\quad \left[1+\left(\frac{a}{1-a}\right)^{\frac{-1}{\eta_m}}\left(\frac{1+\pi^{ss}-\beta}{1+\pi^{ss}}\right)^{\frac{\eta_m-1}{\eta_m}}\right]^{\frac{\eta_m-\eta_c}{1-\eta_m}} \qquad \text{(A.13)}
\end{aligned}
$$

Replacing $(c^{ss})^{-\eta_c}$ with $\left(\frac{c^{ss}}{k^{ss}}\right)^{-\eta_c}\left(\frac{k^{ss}}{n^{ss}}\right)^{-\eta_c}(n^{ss})^{-\eta_c}$, we have:

$$
\begin{aligned}
\frac{(n^{ss})^{\eta_c}}{(1-n^{ss})^{\eta_n}} &= a^{\frac{1-\eta_c}{1-\eta_m}}(1-\alpha)\left(\frac{y^{ss}}{k^{ss}}\right)^{\frac{-\alpha}{1-\alpha}}\left(\frac{c^{ss}}{k^{ss}}\right)^{-\eta_c}\left(\frac{k^{ss}}{n^{ss}}\right)^{-\eta_c} \\
&\quad \left[1+\left(\frac{a}{1-a}\right)^{\frac{-1}{\eta_m}}\left(\frac{1+\pi^{ss}-\beta}{1+\pi^{ss}}\right)^{\frac{\eta_m-1}{\eta_m}}\right]^{\frac{\eta_m-\eta_c}{1-\eta_m}} \qquad \text{(A.14)}
\end{aligned}
$$